Simon Chiche

Inter-University Institute For High Energies

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Research Interests

My research focuses on the study of the most violent phenomena in the Universe. I am strongly involved in the search for ultra-high-energy neutrinos ($E > 10^{17}\,\mathrm{eV}$), which act as unique tracers of cosmic particle acceleration at the highest energies. I am a member of three international collaborations (GRAND, ARA, and RNO-G) dedicated to the radio detection of cosmic rays, gamma rays and neutrinos. My work addresses the challenges of high-energy astroparticle detection with next-generation experiments through the physical modeling of particle cascades and their electromagnetic emission, Monte Carlo simulations and tool development, and data analysis aimed at reconstructing astrophysical observables. The central goal of my research is to connect observational data with astrophysical models, in order to constrain the nature of cosmic accelerators.

- **Thematic:** Cosmic acceleration, ultra-high-energy neutrinos, large-scale radio detection, particle cascades, multi-messenger astronomy
- **Skills:** Analytical modeling of physical processes, numerical simulations and tool development, data analysis and reconstruction of astrophysical signatures

Employment

2024-Now Postdoctoral researcher at **Inter-University Institute For High Energies (IIHE)**, Université Libre de Bruxelles (ULB): Deep cosmic-ray search with RNO-G. Numerical Monte Carlo simulations of cosmic ray radio signals as seen by deep in-ice observers. Analysis of the signal polarization, timing and radiation energy.

Education

10/2020-10/2023 **PhD Thesis at Institut d'Astrophysique de Paris (IAP):**"Looking for ultra-high energy astroparticles in a radio haystack".
Supervised by Kumiko Kotera (IAP) and Olivier Martineau (LPNHE).
- Polarization of inclined-air-shower radio signals for cosmic-ray/neutrino discrimination.

Publication accepted in Astroparticle Physics (arXiv: 2202.06846)
- Numerical simulation tool for a fast and accurate generation of air-shower radio signals.

ICRC proceeding (arXiv: 2107.03206)
- Physical modeling of inclined air-shower emission mechanisms.

Publication accepted in Physical Review Letters (arXiv: 2404.14541)

2018-2020 **Master in Astronomy & Astrophysics at Sorbonne Université**
Graduation with high honours, ranked 1st/36.

2015-2018 **Physics Degree at Sorbonne Université** (Graduation with honours.)

Teaching and mentoring

Students Mentoring

11/2024-Now	PhD thesis, Marine Vilarino: "Search for a diffuse flux of ultra-high energy neutrinos with the Radio Neutrino Observatory in Greenland"
03/2025-05/2025	Bachelor internship, Elodie Curcio: "Simulation of Radio Emissions from Cosmic Ray-Induced Particle Showers for In-Ice Radio Detectors"
2024	Master 2 internship, Nicolas Moller: "Simulating cosmic–ray air shower radio emission for the Askaryan Radio Array"
2022	Master 2 internship, Marion Guelfand: "Lateral distribution of very inclined air-showers"
2021	Master 1 internship, Valentina Richard Romei: "Semi-analytical modeling of the air-showers radio wavefront"

Classes

2024-2026 Teaching assistant at Université Libre de Bruxelles:

- Electrokinetics labs for $\sim$ 20 Bachelor students, 20h.
- Astroparticle Physics tutorials for $\sim$ 10 Master students, 6h.

2021-2022 Teaching assistant at Sorbonne Université:

- Special relativity, tutorials for $\sim$ 20 undergraduate students, 44h.
- Optics and Electromagnetism, practical classes for $\sim$ 20 undergraduate students, 56h.
- Computational Physics, classes and tutorials for $\sim$ 10 undergraduate students, 32h, class preparation.

Awards and fellowships

2026	3-years postdoctoral fellowship, Fonds national de la recherche scientifique FRS-FNRS
2025	Qualified for French permanent assistant professor positions — CNU section 34
2020	PhD fellowship, French Ministry of Higher Education and Research
2015	Award for outstanding academic achievement, Sorbonne Université

Collaborations

GRAND (Giant Radio Array for Neutrino Detection): active member of the analysis working group.
RNO-G (Radio neutrino observatory in Greenland): search for cosmic rays, development of simulation libraries.
ARA (Askaryan Radio Array): data analysis for cosmic ray search, monitoring activities.

Responsibilities

2025	Member of a PhD selection committee for RNO-G
2021-2022	Co-Webmaster of the GRAND collaboration website [web]
2021-2022	Organizer of the weekly GRAND group meetings
2021	Chair and organizer of the PhD annual workshop at IAP

Outreach

10/02/2026	Invited public lecture "À la recherche des accélérateurs cosmiques surpuissants", Forum des Sciences de Villeneuve d'Ascq, France. [web]
2024-2025	Presentation of IIHE radio activities to bachelor students
2024	Masterclass "Particules du cosmos" at IIHE [web]
2022	Science fair at IAP [web]
	Outreach for the "Astronomy Night" at IAP [web]
	Regular conferences for high school students at IAP
2020	Presentation of the Astronomy & Astrophysics Master 2 for CurieOsity's forum at Sorbonne Université
	Outreach for undergraduate interns at IAP
	Presentation of undergraduate studies for high school students at Lycée Arago, 75012 Paris

Talks and seminars

Summary: 23 conference talks (1 plenary invited); 9 invited seminars; 1 poster.

02/09/2026	Contributed talk, TeVPA 2026, Tendo, Japan
10/06/2026	Contributed talk, ARENA2026, Karlsruhe, Germany
26/11/2025	Seminar, AstroParticule et Cosmologie (APC), France
18/11/2025	Seminar, High-energy Journal Club, Brussels, Belgium
06/11/2025	Contributed talk, TeVPA 2025, Valencia, Spain
16/07/2025	Contributed talk, ICRC2025, Geneva, Switzerland
09/09/2024	Contributed talk, RNO-G collaboration meeting, Uppsala, Sweden
14/06/2024	Two contributed talks, ARENA2024, Chicago University, United-States
29/05/2024	Contributed talk, Belgian Physical Society, VUB, Brussels, Belgium
24/05/2024	Contributed talk, RNO-G online collaboration meeting 2024, online
11/03/2024	Contributed talk, Belgian Neutrino meeting, UCLouvain, Louvain-la-Neuve, Belgium
20/09/2023	PhD thesis defense, IAP, Paris, France
02/08/2023	Contributed talk, ICRC2023, Nagoya University, Nagoya, Japan
29/05/2023	Contributed talk, Journées PNHE 2023, IAP, Paris, France
12/05/2023	Contributed talk, IAP Phd annual workshop, IAP, Paris, France
29/03/2023	Contributed talk, Elbereth Conference, Paris, France
17/01/2023	Contributed talk, GRAND collaboration meeting, Radboud University, Nijmegen,
2020-2023	Regular contributions (1 per trimester) to monthly GRAND calls, monthly
27/11/2022	Seminar, High-energy journal club, IAP, Paris, France
	GRAND analysis meetings
18/10/2022	Seminar, Karlsruhe Institute of Technology (KIT) weekly meeting, KIT, Karlsruhe,
07/06/2022	Contributed talk, Acoustic & Radio EeV Neutrino Detection Activities (ARENA),
	Santiago de Compostela, Spain
29/04/2022	Seminar, PhD weekly seminar, IAP, Paris, France
22/04/2022	Contributed talk, IAP PhD annual workshop, IAP, Paris, France
21/03/2022	Invited plenary talk, Rencontres de Moriond, La Thuile, Italy
08/03/2022	Seminar, Vrije Universiteit Brussels (VUB) weekly meeting, VUB, Brussels, Belgium
15/12/2021	Contributed talk, GRAND collaboration meeting, LPNHE, Paris, France
26/10/2021	Contributed talk, Saclay symposium, Institut Pascal, Saclay, France
17/10/2021	Contributed talk, JRJC conference, La Rochelle, France
13/07/2021	Contributed talk, ICRC conference, online
28/05/2021	Seminar, PhD weekly seminar, IAP, Paris, France
04/05/2021	Contributed talk, IAP Phd annual workshop, IAP, Paris, France
09/02/2021	Contributed talk, Elbereth Conference, online
07/12/2020	Poster, Cosmic-ray and neutrino in Paris, online
28/10/2020	Seminar, High-energy journal club, IAP, Paris, France
26/10/2020	Seminar, VUB weekly meeting, VUB, Brussels, Belgium

Publications

Publications listed below include only works to which I contributed actively.

Summary: 6 peer-reviewed publications (4 with major contributions); 1 submitted first-authored papers; 10 conference proceedings.

Publications with major contribution

1. **S. Chiche**, S. Toscano, K. de Vries, "In-ice Radio Signatures of Cosmic Ray Particle Cascades", *Astroparticle Physics* (2026), [2601.06417].

2. **S. Chiche** et al., "Loss of coherence and change in emission physics for radio emission from very inclined cosmic-ray air showers", *Physical Review Letters* (2024), [2404.14541]

3. M. Guelfand, **Simon Chiche** et al.,"Particle content of very inclined air showers for radio signal modeling", *Journal of Cosmology and Astroparticle Physics* (2024), [2310.19612]

4. **S. Chiche**, K. Kotera, O. Martineau, M. Tueros, K. de Vries, "Polarisation signatures in radio for inclined cosmic-ray induced air-shower identification", *Astroparticle Physics* (2022), [2202.06846]

5. F. Durret, **S. Chiche**, C. Lobo, M. Jauzac, "Jellyfish galaxy candidates in MACS J0717.5+3745 and thirty-nine other clusters of the DAFT/FADA and CLASH surveys", *Astronomy & Astrophysics* (2021), [2102.02595]

Publications with minor contribution

1. GRAND collaboration incl. **S. Chiche**, "GRANDlib: A simulation pipeline for the Giant Radio Array for Neutrino Detection (GRAND)", *Computer Physics Communications* (2024), [2408.10926] - Contribution to Section 6 on Shower Generation.

Submitted first-author publications

1. **S. Chiche**, O. Martineau, M. Tueros, K. de Vries, "Radio Morphing: Fast computation of inclined air shower radio emission", submitted to *Astroparticle Physics*, [2601.06418].

Conference Proceedings

1. **S. Chiche**, S. Toscano, K. de Vries, "Radio Signatures of Cosmic-Ray Particle Showers in Deep In-Ice Antennas", *PoS ICRC2025* (2025) 221, [link]

2. **S. Chiche**, for the GRAND collaboration, "GRANDProto300: status, science case, and prospects", *PoS ARENA2024* (2024) 059, [2409.02195]

3. **S. Chiche**, N. Moller, A. Bishop, S. de Kockere, K. de Vries, S. Toscano, "Radio signatures of cosmic-ray showers with deep in-ice antennas", *PoS ARENA2024* (2024) 050, [2409.02185]

4. **S. Chiche**, C. Zhang, K. Kotera, K. de Vries, T. Huege, F. Schlüter, M. Tueros, "New features in the radio-emission of very inclined air-showers", *PoS ICRC2023* (2023) 394, [link]

5. M. Guelfand, **S. Chiche**, et al., "Particle content of very inclined air showers for radio signal modeling", *PoS ICRC2023* (2023) 287, [link]

6. **S. Chiche**, V. Decoene, "Overview of radio experiments for UHE cosmic particles detection", *56th Rencontres de Moriond VHEPU* (2022), [2210.13560]

7. **S. Chiche**, O. Martineau, K. Kotera, M. Tueros, K. de Vries, "Radio-Morphing: a fast, efficient, and accurate tool to compute the radio signals from air-showers", *PoS ICRC2021* (2021) 194, [2202.05886]

8. V. Decoene, O. Martineau, M. Tueros, **S. Chiche**, "A reconstruction procedure for near horizon extensive air showers based on radio signals", *PoS ICRC2021* (2021) 211, [2107.03206]

9. F. Durret, **S. Chiche**, C. Lobo, M. Jauzac, "Hunting for jellyfish galaxies in clusters", *Galaxy Cluster Formation II* (2021) 22, [link]

10. F. Durret, **S. Chiche**, C. Lobo, M. Jauzac, "Jellyfish galaxies in MACS J0717.5+3745 and their link to the cosmic web", *SF2A-2021* (2021) 387-390, [link]

PhD thesis

S. Chiche, "Looking for ultra-high-energy astroparticles in a radio haystack", *Sorbonne Université* (2023), [link]

Other research projects

2nd Semester 2020	**Internship at Institut d'Astrophysique de Paris** Study of ultra-high-energy cosmic-rays from numerical Monte-Carlo simulations modeling the interaction of a cosmic-ray with the Earth's atmosphere. Supervised by Kumiko Kotera (IAP) and Olivier Martineau (LPNHE).
1st Semester 2020	**Lab-Astrophysics project at LERMA** Study of the photodesorption of interstellar ices in order to understand the abundance in gaseous phase of complex molecules in the cold interstellar medium. Supervised by Jean-Hugues Fillion (LERMA).
2nd Semester 2019	**Internship at Institut d'Astrophysique de Paris** Study of jellyfish galaxies from HST images of galaxies. Accepted publication in Astronomy and Astrophysics (arXiv: 2102.02595). Supervised par Florence Durret (IAP).

Other skills

Programming skills: C++, Python, Fortran, Mathematica
Softwares: Siril, DS9, LaTeX, Suite Office
Monte Carlo simulations: ZHaires, CoREAS, GEANT4, FAERIE

Languages: French: native; English: advanced skills; German: Intermediate level; Japanese: Intermediate level

Other responsibilities

2026	**Reviewer** for *JCAP*
2025	**Reviewer** of an *ECRS* conference proceeding
2024-Now	**Station expert** for RNO-G (sanity checks and data quality monitoring)
2024-Now	**Monitoring shifts** for RNO-G